

Campus Marketplace Project Proposal

Verbal Description

Campus Marketplace is a responsive web application that connects college students with businesses operated by other students at or near their university. Many students provide hairstyling, nail services, photography, tutoring, laundry services, clothing, food, and other products or services. However, these businesses are typically advertised through social media, flyers, group chats, and word of mouth. This makes it difficult for customers to discover trustworthy vendors and for student entrepreneurs to reach customers outside their existing social circles. Campus Marketplace will organize these businesses in one searchable, campus-specific platform.

Users will register with a valid `.edu` email address and confirm ownership through an email-verification link. Their account will then be associated with their university. Customers will be able to browse businesses by category, search for specific products or services, filter results by price, availability, and rating, and view vendor profiles. Each profile will include a business description, portfolio photographs, prices, services or products offered, availability, and customer reviews. Customers will also be able to save favorite businesses, submit appointment requests, and review vendors after receiving a service.

Student vendors will be able to create and update business profiles, publish service or product listings, upload portfolio images, set prices and availability, advertise student discounts, and accept or decline booking requests. Administrators will be able to moderate listings, review user reports, and remove inappropriate or fraudulent content. The first version will not process payments or coordinate delivery. Customers and vendors will arrange payment and product collection independently.

Campus Marketplace is comparable to StyleSeat, Yelp, and Etsy. StyleSeat primarily connects customers with beauty professionals, Yelp lists businesses across broad geographic areas, and Etsy focuses on online product sales. Campus Marketplace differs by supporting multiple categories of student-operated businesses and organizing them according to university communities. Its campus-based structure provides convenient discovery while creating greater trust through verified university email addresses.

The application will use React for the front-end interface and Python with FastAPI for the back-end REST API. A PostgreSQL database will store user accounts, vendor profiles, listings, booking requests, and reviews. AWS Amplify will host the React application, while an appropriate cloud service will host the back-end API and database. GitHub will support version control and team collaboration. Development will require internet-connected computers, modern browsers, and software-development and testing tools. Vendors will maintain their own listings, while administrators and developers will manage database backups, security updates, reported content, and future application improvements.

Justification

This project is appropriate for the Senior Project Laboratory because it applies knowledge gained throughout the computer science curriculum, including web development, database design, software engineering, user-interface design, authentication, API development, testing, and version control. It will also extend the team's knowledge through the implementation of email verification, role-based permissions, campus-specific discovery, image storage, search filters, booking workflows, and secure data management.

The project offers an appropriate level of technical difficulty because it requires several connected components while remaining achievable within one semester. The team can prioritize a minimum viable product consisting of account registration, vendor profiles, listings, search, booking requests, and reviews. Features such as integrated payments, delivery tracking, and real-time messaging can be reserved for future development. This scope allows the team to produce a complete and usable application without attempting an unrealistic number of features.

The team has the programming experience and development tools necessary to design, implement, and test the system. The project is also relevant and interesting because student entrepreneurs already operate across college campuses but lack a centralized way to advertise their businesses and build a customer base. Campus Marketplace addresses a recognizable student problem, supports student entrepreneurship, and has the potential to expand to additional universities after the initial version is completed.